Fiscal Year:	FY 2015	**Task Last Updated:**	FY 05/18/2015
PI Name:	Nickerson, Cheryl A Ph.D.		
Project Title:	Investigation of Host-Pathogen Interactions, Conserved Cellular Responses, and Countermeasure Efficacy During Spaceflight using the Human Surrogate Model Caenorhabditis elegans		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology (3) Animal Biology: Invertebrate		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology (2) Immunology		
Space Biology Special Category:	None		
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Organization Name:	Arizona State University		
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City:	Tempe	**State:**	AZ
Zip Code:	85287-5401	**Congressional District:**	9
Comments:	NOTE PI moved from Tulane University to Arizona State University in 2006.		
Project Type:	Flight	**Solicitation / Funding Source:**	2009 Space Life Sciences-Flight NNH09ZTT002N
Start Date:	07/13/2010	**End Date:**	09/30/2017
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Taylor, Elizabeth	**Contact Phone:**	650.604.1783
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: Extended to 09/30/2017 per F. Hernandez/ARC (Ed., 2/21/17) NOTE: Extended to 2/28/2017 per PI (Ed., 3/1/2016) NOTE: End date is now 2/29/2016 per NSSC information (Ed., 2/25/15) NOTE: End date is now 2/28/2015 per NSSC information (Ed., 3/13/14) NOTE: End date is now 7/12/2014 per NSSC information (Ed., 2/1/2013)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Alverdy, John (University of Chicago) Ott, C Mark (NASA Johnson Space Center)		
Grant/Contract No.:	NNX10AO52G		

Performance Goal No.:	
Performance Goal Text:	
Task Description:	Changes that occur to both the host immune system and pathogenesis of microbes during spaceflight represent a formidable challenge to the successful transition from short-to-long-duration spaceflight. This is a critical concern, since a) the immune system of the crew is dysfunctional during spaceflight, b) spaceflight uniquely increases the virulence and globally alters gene expression of the human pathogen Salmonella typhimurium when the bacteria are recovered post-flight and subsequently used to infect ground-based animal models, c) spaceflight-induced increases in S. typhimurium virulence are regulated by media ion composition, d) phosphate ion is sufficient to alter related pathogenesis responses in a ground-based spaceflight analogue model, and e) the evolutionarily conserved bacterial RNA chaperone protein, Hfq, serves as a master molecular regulator of many of these responses. While spaceflight has been shown to induce changes that can independently affect the host or the pathogen in a manner that is directly relevant to the development of infectious disease during spaceflight, none of these studies have been done when both the host and pathogen are simultaneously exposed to the spaceflight environment. To address this knowledge gap, we propose to challenge the human surrogate model Caenorhabditis elegans with S. typhimurium during spaceflight to study the role of the intestine in host-pathogen interactions in real-time, define virulence mechanisms, identify evolutionarily conserved responses, and test novel therapeutic strategies to prevent infectious disease. We hypothesize that the combination of a spaceflight-induced increase in virulence of the pathogen coupled with a blunted immune response of the host will result in a synergistic effect on the host-pathogen interaction such that the risk of infectious disease during spaceflight is exacerbated. We further propose that these responses represent evolutionarily conserved mechanisms at the level of both host and pathogen that are dependent on media ion composition, specifically the local concentration of phosphate. See also: http://www.nasa.gov/
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research will enrich life on Earth through the use of space technology and the application of biomedical knowledge. Specifically, this study will utilize the microgravity spaceflight platform 1) to broaden our knowledge of the host-pathogen interaction that leads to infectious disease, and 2) for the development of new therapeutic strategies to combat infectious disease for the general public.
Task Progress:	Previously published results from our lab showed that spaceflight uniquely increases the virulence and globally alters gene expression of the major human pathogen Salmonella typhimurium when the bacteria are recovered post-flight and subsequently used to infect laboratory-based animal models. In our efforts to determine the mechanism(s) behind the spaceflight-induced responses in Salmonella, we discovered that a) spaceflight-induced increases in Salmonella virulence are regulated by media ion composition, b) phosphate ion is sufficient to alter related pathogenesis responses in a spaceflight analogue model, c) the Hfq protein serves as a master molecular regulator of many of these responses in Salmonella, and d) Hfq also acts as a spaceflight-induced regulator acting across bacterial species, which to our knowledge, is the first report of such a master response regulator of the spaceflight-induced stimulon in bacteria. Hfq is a highly conserved bacterial RNA chaperone protein that binds to small non-coding regulatory RNAs thereby facilitating their association with mRNAs, the result of which plays a diverse role in prokaryotic gene expression (including regulation of ion response pathways), virulence, and physiology in response to stress. While spaceflight has been shown to induce changes that can independently affect the host or the pathogen in a manner that is directly relevant to the development of infectious disease during spaceflight, no studies have been published that meaningfully document the outcome when both the host and pathogen are simultaneously exposed to the spaceflight environment. Furthermore, the eukaryotic organism and human surrogate infection model, C. elegans, contains structural and functional homologues of Hfq proteins, which are a family of evolutionarily conserved RNA-binding proteins that are critical for immune function, cell differentiation/function, and responses to cellular stress. We hypothesize that the combination of a spaceflight-induced increase in virulence of the pathogen coupled with a blunted immune response of the host will result in a synergistic effect on the host-pathogen interaction such that the risk of infectious disease during spaceflight is exacerbated and may uncover common conserved responses in the host and pathogen that can be targeted for countermeasure development. We have successfully completed the objectives below in support of the spaceflight investigation (operationally called MICRO-5/PHOENIX) which launched aboard SpaceX-5 to the International Space Station on January 10, 2015: - Successful completion of our EVT (Engineering Verification Test) in early July of 2014. EVT operations were initiated in early June and all operations were performed smoothly and successfully. Video analysis and RNA integrity assessment were completed in early July and proved to be of the high integrity and quality needed. - The science team arrived at the Kennedy Space Center (KSC) on November 4, 2014 for initial laboratory set up, media preparations, and experimental setup in order to meet the initial December 9th 2014 launch attempt. *ACTUAL LAUNCH SLIPPED TO JANUARY 10, 2015. - Operations on orbit as well as the coordinating ground controls (the latter performed by the science team at KSC) occurred one day later than anticipated due to the ammonia leak alarm and emergency evacuation of the crew. This delay (which fortunately turned out to be a false alarm) still allowed operations to begin within the timeline required to ensure experimental integrity. All operations were completed within the Microgravity Science Glovebox. The science team was able to view the on orbit operations in the Experiment Monitoring Area (EMA) at Kennedy Space Center. Viewing showed that astronaut Terry Virts followed step-by-step instructions from BioServe to complete each operation and utilized sterile technique while handling the hardware as trained by BioServe during the astronaut training sessions. The science team and BioServe were able to communicate with the crew in real time. - Post-Flight sample return occurred in segments. Ground Control samples arrived to the PI (Principal Investigator) lab by hand carry from KSC personnel. The flight samples arrived at Long Beach Airport in CA on 2-11-15 and were met by BioServe and the science team at that time. All early destow samples and hardware were inspected prior to handover to the science team for hand carry back to the PI lab. Remaining samples were returned to BioServe in Boulder, Colorado, about 6 weeks after SpaceX-5 return. The science team traveled to Boulder to retrieve those samples and shipped them back to the PI lab. - Samples are currently being processed*. *(Note: Completion of sample analysis, including gene expression profiling,

	is awaiting necessary fund transfers to support this work from NASA to the PI). Initial testing of the sample contents has indicated that the infection was completed correctly with the pathogen and control bacteria types placed into the correct wells and contained within the hardware both on orbit and on the ground. There were no biocompatibility anomalies to report. Both video analysis and post-flight inspections demonstrated that the bacterial cultures were activated and grew on orbit. We also observed that the nematodes were highly viable at the time of infection and were successfully activated out of the dauer state using the control (E. coli) and infection (Salmonella) bacterial cultures. Significal Media Coverage events and Awards are listed in the below Bibliography. Presentations were made at the following: Keynote Speaker, Leading Through Collaboration, International Society for Medical Publication Professionals (ISMPP), Arlington, VA April 7-9, 2014. Invited speaker, Phoenix Rotary Club, Phoenix Country Club, Phoenix, AZ, Oct 3, 2014. Invited keynote speaker, President's Symposium, American Society for Space and Gravitational Research, Pasadena, CA, Oct 22, 2014. Invited panel participant, Arizona Strategic Futures Initiative event, ASU Fulton Center, Nov 3, 2014.
Bibliography Type:	Description: (Last Updated: 07/02/2025)
Articles in Peer-reviewed Journals	Crabbé A, Liu Y, Sarker SF, Bonenfant NR, Barrila J, Borg ZD, Lee JJ, Weiss DJ, Nickerson CA. "Recellularization of decellularized lung scaffolds is enhanced by dynamic suspension culture." PLoS One. 2015 May 11;10(5):e0126846. eCollection 2015. http://dx.doi.org/10.1371/journal.pone.0126846 ; PubMed PMID: 25962111; PubMed Central PMCID: PMC4427280 , May-2015
Awards	Nickerson C. "Founding Editor-in-Chief of Nature Publishing Group journal, npj Microgravity. Dedicated to publishing the most important scientific advances in the life sciences, physical sciences, and engineering fields that are facilitated by spaceflight and analogue platforms. Journal launch, May 2015. http://www.nature.com/npjmgrav/ " May-2015
Awards	Nickerson C. "Invited Committee Member, Session Lead, Institute of Medicine, National Academy of Sciences review of NASA's Evidence Reports of Human Health Risks, May – November, 2014." Nov-2014
Books/Book Chapters	Nickerson CA, Ott CM, Pellis N. (Eds.) "Effect of Spaceflight and Analogue Culture on Human and Microbial Cells: Novel Insights into Disease Mechanisms." New York : Kluwer Academic/Plenum Press, 2015. In press as of May 2015., May-2015
Significant Media Coverage	Space Station Live: Micro-5. Commentator Brandi Dean. "NASA Commentator Brandi Dean talks with Dr. Cheryl Nickerson of Arizona State University about her International Space Station experiment Micro-5, which is designed to improve our understanding of the risk of infections faced by long-duration space travelers. The experiment was delivered by the recent Dragon cargo ship, which will return experiment samples to Earth for analysis." NASA featured video on YouTube, January 26, 2015. http://www.vidinfo.org/video/6827075/space-station-live-micro-5 ; https://www.youtube.com/watch?v=df1ck6RjdYc ; accessed 5/18/15. , Jan-2015
Significant Media Coverage	Harth R. "Mysteries of Infection Charted in Spacebound Worm." Article about Dr. Nickerson's Micro-5 experiment onboard ISS. January 15, 2015. http://www.biodesign.asu.edu/news/mysteries-of-infection-charted-in-spacebound-worm ; accessed 5/18/15., Jan-2015
Significant Media Coverage	Invited Speaker and Panel Member, SpaceX CRS-5 and International Space Station Research and Technology Briefing. "SpaceX CRS-5 and International Space Station Research and Technology Briefing." Pre-flight Press Conference, Kennedy Space Center, Cape Canaveral, FL, Jan 5, 2015., Jan-2015
Significant Media Coverage	Google Science Fair Hangout interview by Scientific American editor in chief, Mariette DeChristina. "Google Science Fair Hangout interview, about the Micro-5 flight experiment, of Dr. Nickerson, in collaboration with Nature Publishing Group and Scientific American." Google Science Fair Hangout interview of Nickerson by Scientific American editor in chief, Mariette DeChristina, October 10, 2014. https://plus.google.com/events/ct270lh41it3bj356aajeu33q0c ; accessed 5/18/15., Oct-2014